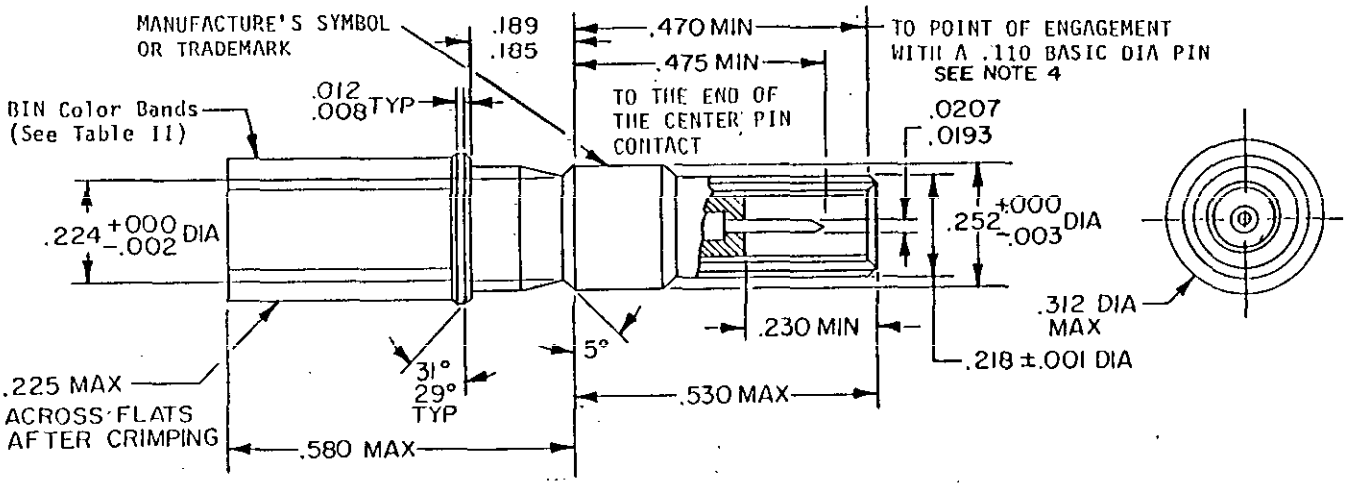
MILITARY SPECIFICATION SHEET

CONTACTS, ELECTRICAL CONNECTOR, CONCENTRIC TWINAX, PIN, SHIELDED, SIZE 8 (FOR MIL-C-28840)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the contacts described herein shall consist of this specification and the latest issue of MIL-C-39029.

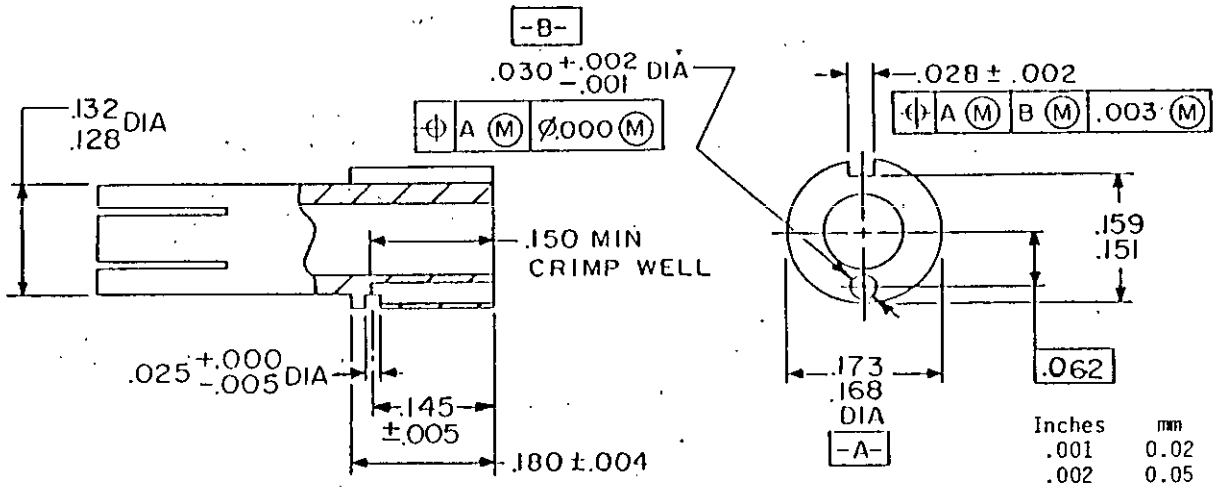


Inches	mm	Inches	mm	Inches	mm
.001	0.02	.0207	0.52	.230	5.84
.002	0.05	.110	2.79	.252	6.40
.003	0.07	.185	4.69	.312	7.92
.008	0.20	.189	4.80	.470	11.93
.012	0.30	.218	5.53	.475	12.06
.0193	0.49	.224	5.68	.530	13.46
		.225	5.71	.580	14.73

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions shown apply after plating.
4. Point at which a square ended pin of the same basic diameter as the mating contact first engages the intermediate contact spring. Provision for clearance hole shall be provided for the test pin.

FIGURE 1. Concentric twinax contact.

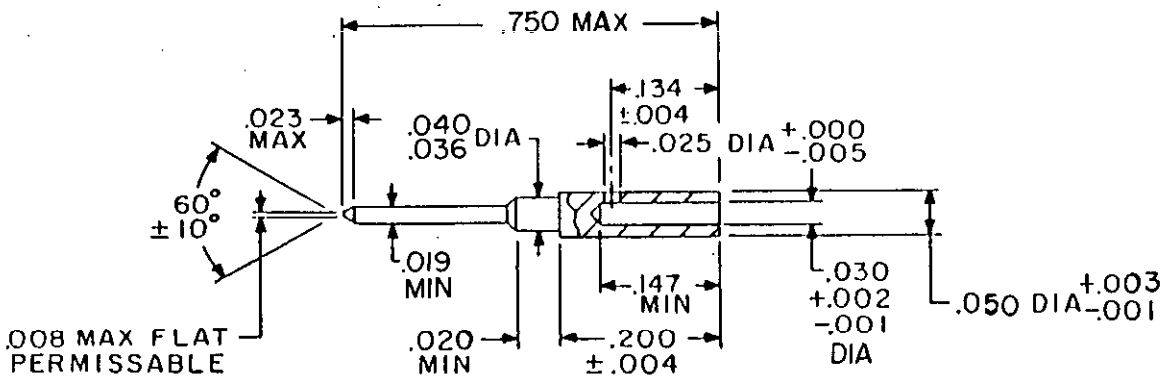
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:46Z
Check the source to verify that this is the current version before use.



Inches	mm
.001	0.02
.002	0.05
.003	0.07
.004	0.10
.005	0.12
.025	0.63
.028	0.71
.030	0.76
.062	1.57
.128	3.25
.132	3.35
.145	3.68
.150	3.81
.151	3.83
.159	4.03
.168	4.26
.173	4.39
.180	4.57

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions shown apply after plating.

FIGURE 2. Intermediate socket contact.



Inches	mm	Inches	mm	Inches	mm
.001	0.03	.019	0.48	.040	1.02
.002	0.05	.020	0.51	.050	1.27
.003	0.08	.023	0.58	.134	3.40
.004	0.10	.025	0.64	.147	3.73
.005	0.13	.030	0.76	.200	5.08
		.036	0.91	.750	19.05

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimensions shown apply after plating.

FIGURE 3. Center pin contact.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:46Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Qualification:

Contacts shall comply with the reliability assurance provisions of MIL-STD-790 as specified in MIL-C-28840.

Design and construction:

Dimensions and configurations: See figures 1 through 3 and table I.

TABLE I. Design characteristics.

BIN code	Color bands			Contact cavity size	Cable accommodated	Type	Class
	1st	2nd	3rd				
556	Green	Green	Blue	8	M17/176-00002	D	B

Assembly procedure: Manufacturer's recommended assembly instructions shall be shipped with unit package.

Electrical:

Low signal level contact resistance (center and intermediate contacts only): See table II.

Contact resistance: See table III.

Frequency: 0 to 20 MHz (operating frequency range).

Voltage rating: 500 volts rms maximum; working voltage at sea level, 125 volts rms maximum at 70,000 feet.

TABLE II. Low signal level contact, resistance (center and intermediate contacts only).

BIN code	Cable accommodated	Maximum contact resistance (milliohms)	
		Initial	After conditioning
556	M17/176-00002	55	66

TABLE III. Contact resistance.

BIN code	Contact	Cable accommodated	Test current (amperes)	Maximum voltage drop (millivolts)		
				+25°C, +3°C 0°C		+175°C, +3°C, 0°C
				Initial	After conditioning	After conditioning
556	Center	M17/176-00002	1.0	55	66	94
556	Intermediate	M17/176-00002	1.0	55	66	94
556	Outer	M17/176-00002	12.0	75	90	128

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:46Z
Check the source to verify that this is the current version before use.

Dielectric withstanding voltage: Shall be as specified in table IV.

TABLE IV. Dielectric withstanding voltage.

Contacts	Altitude	Test voltages ac rms
Center to intermediate	Sea level	1,000
Intermediate to outer	Sea level	500

Mating contact: Shall be in accordance with MIL-C-39029/105.

Mechanical:

Contact engagement and separation force (socket contacts only): The engagement depth shall be a minimum of 0.7 of the minimum socket bored depth. The test pins shall be in accordance with MS3197, except the diameters shall be as specified in table V. Provision for clearance hole on outer contact. Test pins shall be provided.

TABLE V. Contact engagement and separation force.

Test pin diameter (inch)	Minimum separation force (ounces)		Maximum engagement force (ounces)		Maximum average engagement force
	Initial	After conditioning	Initial	After conditioning	
.1050 +.0000 -.0001	NA	NA	18	22	NA
.1110 +.0001 -.0000	0.5	0.4	NA	NA	NA

Crimp tensile strength (center, intermediate, and outer contact crimp joint): See table VI.

TABLE VI. Crimp tensile strength (at ambient).

BIN code	Cable accommodated	Axial load (pounds, minimum)		
		Center contact	Intermediate contact	Outer contact
556	M17/176-00002	8	8	25

Vibration: Method 2005 of MIL-STD-1344, test condition V. The following details shall apply:

a. Use the vibration envelope shown on figure 4.

b. Vibration: To be conducted at standard test conditions.

c. Duration: Eight hours in the longitudinal direction and 8 hours in a perpendicular direction (16 hours).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:46Z
Check the source to verify that this is the current version before use.

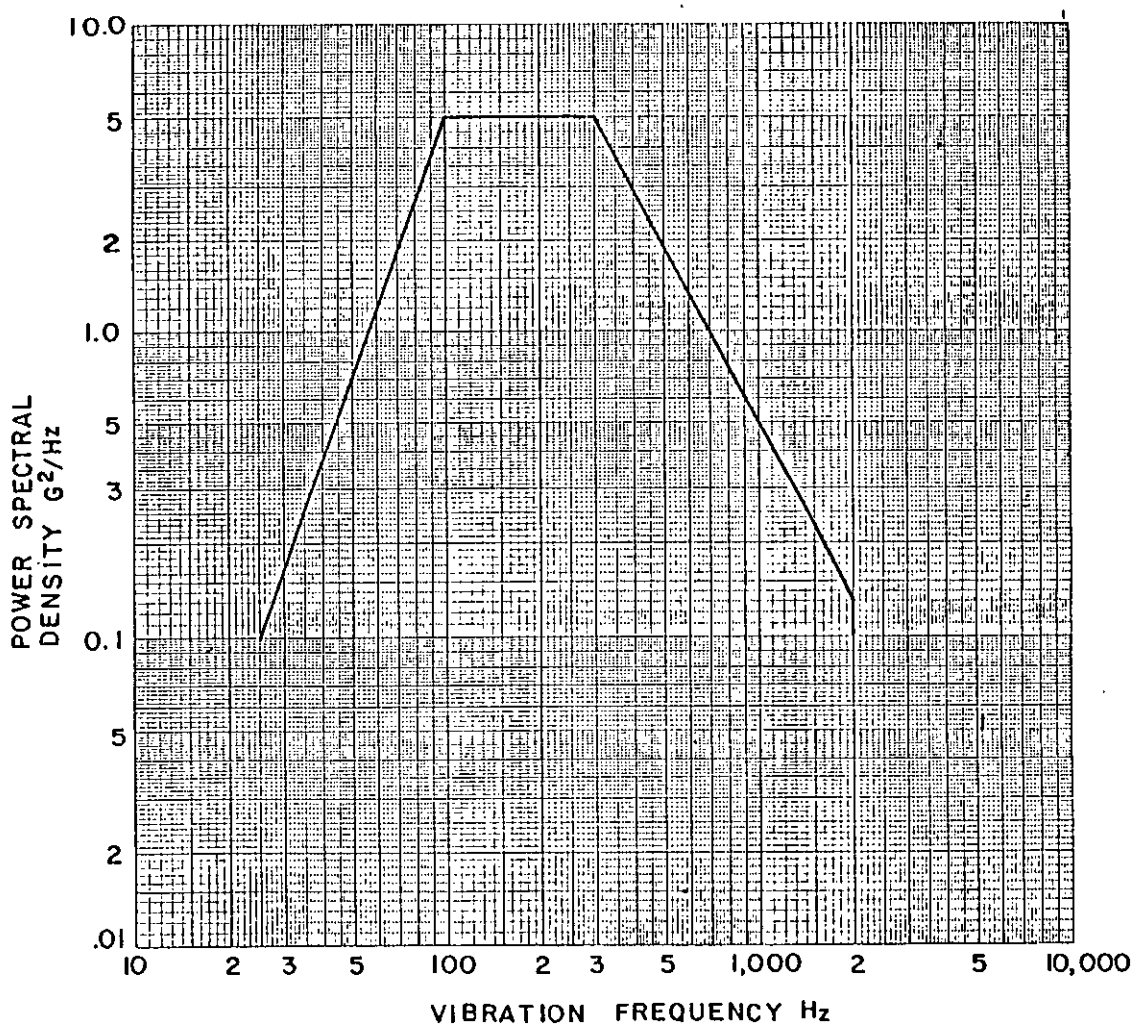


FIGURE 4. Vibration envelope.

High impact shock: All contacts shall be wired in a series circuit with 100 milliampers maximum current flow through the series circuit during shock. Contacts shall be monitored for any discontinuities. A detector capable of detecting all discontinuities in excess of 1 microsecond shall be used. Wired and mated contacts shall be subjected to the test specified in MIL-S-901, grade A with the following modifications and additions. Mounting fixture shall be in accordance with MIL-S-901. The cable or wire bundle shall be supported on a stationary frame to provide a free flexing cable length between the frame and fixture of not less than 36 inches (914.4 mm).

a. Coupling: Coupled by normal mounting means.

b. Test condition A: The wire bundle shall be provided with straight, open frame, strain relief accessory hardware.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:46Z
Check the source to verify that this is the current version before use.

Tools: Shall be as specified in tables VII and VIII.

TABLE VII. Tools (installing and removal).

BIN code	Installing tool	Removal tool
556	M81969/17-06	M81969/19-03

TABLE VIII. Tools (crimping).

BIN code	Center contact tooling		Intermediate contact tooling		Outer crimp sleeve tooling	
	Basic crimping tool	Contact positioner 1/	Basic crimping tool	Die part number 1/	Basic crimping tool	Die part number 1/
556	M22520/2-01	K838	M22520/5-01	Y779	M22520/5-01	Y779

1/ Daniels Manufacturing Corporation part number is specified. Tools may be provided by other vendors if they are determined to be equivalent by the qualifying activity.

Environmental:

Temperature rating: -55°C to +200°C.

Humidity: In accordance with MIL-STD-1344, method 1002, type II.

Part number: See table IX.

TABLE IX. Part number.

BIN code	Part number
556	M39029/104-556

QPL evaluating activity: Naval Weapons Support Center, Crane, IN 47522.

Custodians:
Navy - EC
Army - CR
Air Force - 85

User activities:
Navy - OS
Air Force - 11

Review activities:
Navy - AS
Marine Corp - MC
DLA - ES

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5935-3538-01)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T18:46Z
Check the source to verify that this is the current version before use.